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Resistive, Inductive & Capacitive Transducers

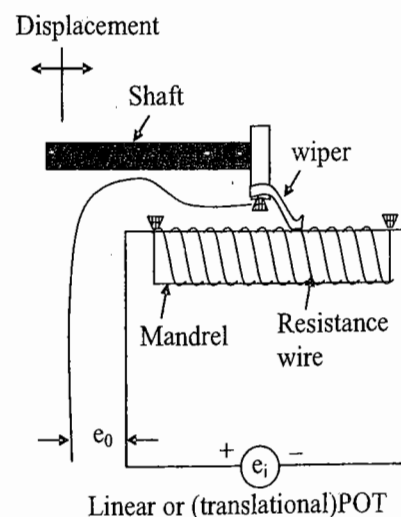
- It is generally seen that methods which involve the measurement of change in resistance are preferred to those employing other principles.
- This is because both alternating as well as direct currents and voltage are suitable for resistance measurements.
- The resistance of a metal conductor is expressed by a simple equation that involves a few quantities.
- The relation is $R = \frac{\rho L}{A}$
- Any method of varying one of the quantities involved in the above relationship can be design basis of an **Electrical Resistance Transducer**.

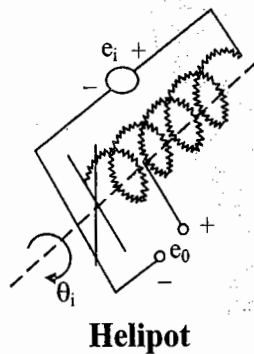
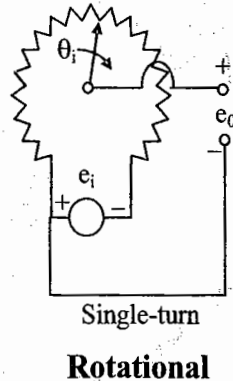
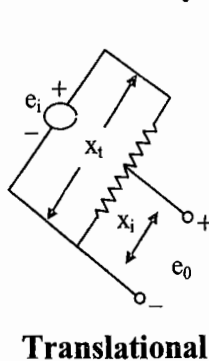
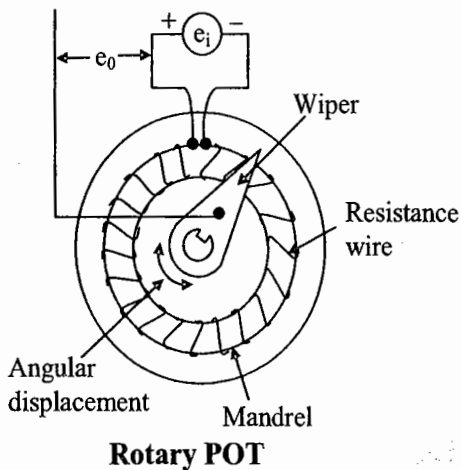
The following table indicates the non-electrical quantities that can be measured by the resistance transducer.

Effect	As a primary transducer	Through the use of secondary transducers
1. Change in length	Linear and angular displacements, thickness	Temperature, pressure, weight, force, torque, level, density, fluid flow rate, viscosity, velocity, acceleration and altitude of vehicles in space.
2. Change in length and area of cross section	High pressure strain	All the above quantities except velocity and altitude
3. Change in resistivity	Temperature	Thermal conductivity, gas composition, fluid flow rate, wind velocity and direction, low pressure and vacuum and infrared radiation.

Potentiometer

- Basically a potentiometer or simply a POT (a resistive potentiometer used for the purpose of voltage division is called a POT) consists of a resistive element provided with a sliding contact.
- This sliding contact is called a wiper.
- The motion of the sliding contact may be translatory or rotational.
- Some POTs use the combination of the two motions, i.e., translational as well as rotational.
- These POTs have their resistive element in the form of a helix and therefore they are called "**Helipots**".





- A full single turn potentiometer may provide accurate measurements upto 357° .
- Multiturn potentiometers may measure upto 3500° of rotation through use of helipots.

Power Rating of Potentiometers

- The potentiometers are designed with a definite power rating which is related directly to their heat dissipating capacity.
- Normally available pots are of single turn with a diameter of 50 mm with a wide range of ohmic values ranging from 100Ω to $10K\Omega$ in steps of 100Ω .

- Power rating is typically 5W at an ambient temperature of 21°C .
- Since power $P = e_i^2 / R_P$, the maximum input excitation voltage that can be used is $(e_i)_{\max} = \sqrt{PR_P}$

Linearity and Sensitivity of Pots:

- Linearity implies resistance of POTs should be low.
- To achieve good sensitivity the output voltage e_o should be high which in turn requires a high input voltage e_i and this calls for a high value of resistance of POT.
- Therefore linearity and sensitivity are two conflicting requirements.

Materials Used for potentiometers:

Based on the materials used POTs may be classified as (i) Wire Wound

(ii) Non – wire wound

- Wire wound potentiometers :
platinum, nickel chromium, nickel copper
- Non-wire wound potentiometers :
cermet, hot moulded carbon, carbon film, thin metal film.

Advantages:

- Inexpensive
- Simple to operate
- Electrical efficiency is high

Disadvantage:

- Linear POT require a large force to move their sliding contacts.

Example 1:

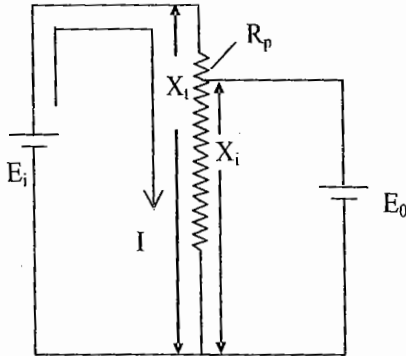
An ideal or unloaded potentiometer used as a displacement transducer has a stroke of 100mm and its resistance over this length is 1000Ω . Make calculation for the power dissipated by the coil. It



may be presumed that the transducer has an overall sensitivity of 0.1 V/mm.

Ans: 0.1 watt

Sol:



Given

$$X_t = 100 \text{ mm}$$

$$R_p = 1000 \Omega$$

$$\text{Sensitivity (S)} = \frac{0.1 \text{ V}}{\text{mm}} = \frac{E_0}{X_i}$$

$$E_0 = 0.1 \text{ V} \rightarrow X_i = 1 \text{ mm}$$

$$\text{When } X_i = X_t = 100 \text{ mm}$$

$$E_0 = E_i = 10 \text{ V}$$

$$I = \frac{E_i}{R_p}$$

$$= \frac{10 \text{ V}}{1000 \Omega} = 0.01 \text{ Amp}$$

So power dissipated by the coil is

$$\begin{aligned} & (I)^2 \times R_p \\ &= (0.01)^2 \times 1000 \Omega \\ &= 0.1 \text{ watt} \end{aligned}$$

Strain Gauge

Strain gauges are used for measurement of strain and associated stress and these can be used as secondary transducers in many applications. They are based on the principle of piezoresistive effect.

Piezoresistive effect:

If a metal conductor is stretched or compressed, its resistance changes on account of the fact that both length and diameter of conductor change. Also there is a change in value

of resistivity of the conductor when it is strained. This property is known as piezoresistive effect.

Theory of Strain Gauges:-

Let us consider a strain gauge made of circular wire.

The wire has the dimensions,

$$\text{Length} = L$$

$$\text{Area} = A$$

Diameter = D before being strained.

The material of the wire has a resistivity ρ .

$\therefore$ Resistance of unstrained gauge, $R = \rho L/A$.

Let a tensile stress s be applied to the wire.

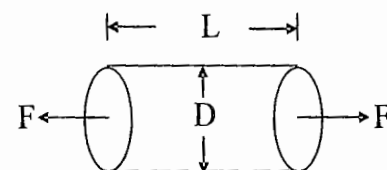
This produces a positive strain causing the length to increase and area to decrease. Thus when the wire is strained there are changes in its dimensions.

Let ΔL = change in length

ΔA = change in area

ΔD = change in diameter

ΔR = change in resistance



Gauge factor:

It is defined as the ratio of per unit change in resistance to per unit change in length.

$$\text{Gauge factor, } G_f = \frac{\Delta R/R}{\Delta L/L}$$

$$\text{or } \frac{\Delta R}{R} = G_f \frac{\Delta L}{L}$$

$$= G_f \times \epsilon$$

$$\epsilon = \text{strain} = \Delta L/L$$

Dividing equation (1) with $\Delta L/L$ we get



$$G_f = 1 + 2\nu + \frac{\Delta\rho/\rho}{\epsilon}$$

$$= 1 + 2\nu + \frac{\Delta\rho/\rho}{\epsilon}$$

Resistance change due to change in length to change in area of piezoresistive effect.

If the change in the value of resistivity of a material when strained is neglected, the gauge factor is $G_f = 1 + 2\nu$

Characteristics of strain gauges:

- Element should be of low resistance temperature coefficient.
- Should have high value of gauge factor
- Should have linear characteristics
- Should have very high resistance.
- Should have very good frequency response.

Types of Strain Gauges:

- Unbonded metal strain gauges
- Bonded metal wire strain gauges
- Bonded metal foil strain gauges
- Semiconductor strain gauges
- Diffused metal strain gauges

Temperature compensation:

The resistive type strain gauges are sensitive to temperature. Therefore it becomes necessary to account for variations in the strain gauge resistance which occurs on account of temperature changes.

Temperature compensation may be provided by

- Use of adjacent arm balancing or compensating gauge.

Use of Dummy gauge

One of the ways in which temperature error can be eliminated by using adjacent arm compensating gauge is to use a dummy gauge in the adjacent arm. This arrangement is shown. Gauge 1 is installed on the test specimen gauge is installed on a like piece of material and is not subjected to any strain.

The gauges installed on the test temperature. A gauge is called dummy gauge in case it is not subjected to any strain. Active gauge is one which is subjected to strain.

Initially when the bridge is balanced,

$$\frac{R_1}{R_3} = \frac{R_2}{R_4}$$

Supposing a change in temperature occurs, the resistance R_1 & R_3 change by an amount ΔR_1 and ΔR_3 respectively.

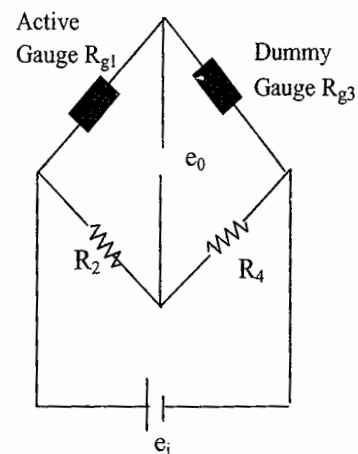
$$\text{Hence for balance, } \frac{R_1 + \Delta R_1}{R_3 + \Delta R_3} = \frac{R_2}{R_4}$$

$$\text{or } \frac{R_4}{R_2} (R_1 + \Delta R_1) = (R_3 + \Delta R_3)$$

$$\text{or } \frac{R_4}{R_2} R_1 + \frac{R_4}{R_2} \Delta R_1 = R_3 + \Delta R_3$$

$$\text{But } \frac{R_4}{R_3} R_1 = R_3 \therefore \frac{R_4}{R_2} \Delta R_1 = \Delta R_3$$

Suppose $R_4 = R_2$ This requires that $\Delta R_1 = \Delta R_3$



It means that for the bridge to remain insensitive to variations in temperature the gauges R_1 and R_2 should have their resistances change by equal amount when subjected to variation in temperature. Therefore the active gauge R_1 and the dummy gauge R_3 should be identical.



The use of dummy gauge for temperature compensation is simple and effective and should be employed whenever possible.

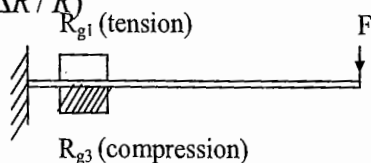
- **Use of two active gauges in adjacent arms**

In certain applications, where equal and opposite strains are known to exist, it is possible to attach two similar gauges in such a way that one gauge experience a positive strain and the other a negative strain. Thus instead of having an arrangement wherein one gauge acts as the active gauge and the other as the dummy gauge, we have now an arrangement wherein both the gauges are active gauges.

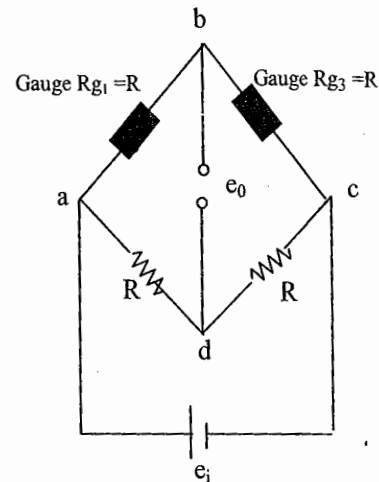
Fig shows the two gauges mounted on a cantilever. The gauge R_{g1} is on top of the cantilever and hence experiences tension or a positive strain. The R_{g3} is at the bottom surface

of the cantilever and hence experiences a compression or a negative strain.

- The bridge arrangement for the two gauges is shown in Fig., There are two active gauges in the 4 arm bridge and hence it is called a Half Bridge.
- The temperature effects are cancelled out by having $R_2 = R_3$ and using two identical gauges in the opposite arms of the bridge.
- When no strain is applied both points b and d are at the same potential, $e_i/2$ and the value of output voltage $e_0=0$.
- When the arrangement shown in Fig is subjected to strain, the resistance of gauge 1 increases and that of gauges R_{g3} decreases. Resistance of gauge R_{g1} when strained is $R(1 + \Delta R/R)$



Two gauges used for measurement of strain



Adjacent arm compensation using two active gauges

Resistance of gauge R_{g3} when strained is $R(1 - \Delta R/R)$

Now $R_2 = R_4 = R \therefore$ Potential of point d is $= e_i/2$

$\therefore$ Potential of point

$$b = \frac{R(1 + \Delta R/R)}{R(1 + \Delta R/R) + R(1 - \Delta R/R)} \times e_i$$

$$= \frac{1 + \Delta R/R}{2} e_i$$

$\therefore$ Change in output voltage when strain applied is,

$$\Delta e_0 = \frac{1 + \Delta R/R}{2} e_i - \frac{e_i}{2} = \frac{\Delta R/R}{2} e_i = \frac{G_f \epsilon}{2} e_i$$

Thus the output voltage from a half bridge is twice that from a quarter bridge and therefore the sensitivity is doubled. In addition, the temperature effects are cancelled. The gauge, sensitivity of a half bridge is

$$S_g = 2kR_g G_f$$

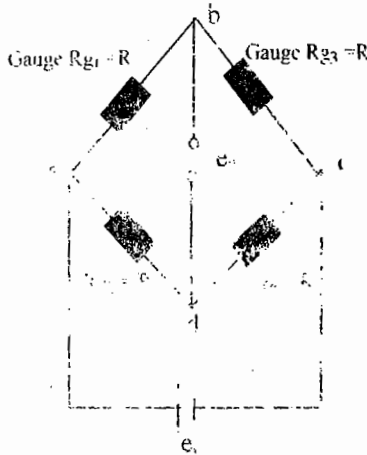
- **Use of four active gauges**

- Fig shows a cantilever using 4 strain gauges for the measurement of strain. All the four gauges are similar and have equal resistance when strained

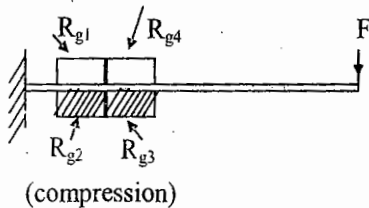
$$R_{g1} = R_{g2} = R_{g3} = R_{g4} = R.$$

- These gauges are connected in the arms of a Wheatstone bridge as shown in fig., since the bridge has 4 active gauges with one gauge in

each of the four arms it is called a **Full Bridge**.



Bridge circuit for measurement of strain four using active gauges



- When no strain is applied the potential of points b and d are both equal to $e_i/2$ and hence the output voltage $e_o = 0$.

When strained, the resistance of various gauges are:

For R_{g1} and R_{g4} : $R(1 + \Delta R/R)$ and for R_{g2} and R_{g3} : $R(1 - \Delta R/R)$

Potential of b when strain is applied

$$\frac{R(1 + \Delta R/R)}{R(1 + \Delta R/R) + R(1 - \Delta R/R)} e_i = \frac{1 + \Delta R/R}{2} e_i$$

Potential of d when strain is applied

$$\begin{aligned} &= \frac{R(1 - \Delta R/R)}{R(1 - \Delta R/R) + R(1 + \Delta R/R)} e_i \\ &= \frac{1 - \Delta R/R}{2} e_i \end{aligned}$$

Therefore change in output voltage

$$\begin{aligned} \Delta e_o &= \frac{1 + \Delta R/R}{2} e_i - \frac{1 - \Delta R/R}{2} e_i = (\Delta R/R) e_i \\ &= G_f \epsilon e_i \end{aligned}$$

Four active-arm bridges are extensively used when strain gauges are used as secondary transducers to give maximum sensitivity combined with full temperature compensation. The effect of increasing the number of active gauges is the same if a low impedance detector is used.

The gauge sensitivity of a full bridge

$$S_g = 4K R_g G_f$$

- Self compensation
- Use of special external control circuitry.

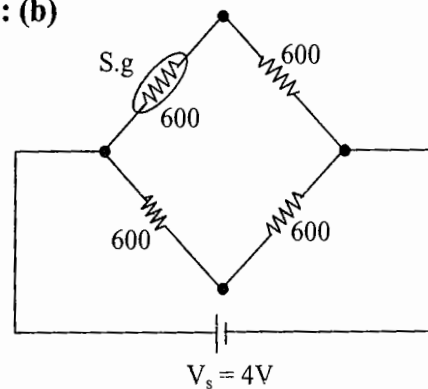
Example 2:

A strain gauge has a nominal resistance of 600Ω and a gauge factor of 2.5. The strain gauge is connected in a dc bridge with three other resistances of 600Ω each. The bridge is excited by a 4 V battery. If the strain gauge is subjected to a strain of $100 \mu\text{m}/\text{m}$, the magnitude of the bridge output will be

- (a) 0 V
- (b) $250 \mu\text{V}$
- (c) $500 \mu\text{V}$
- (d) $750 \mu\text{V}$

Ans: (b)

Sol:



Given

$R = 600$ (strain gauge resistance)

$G.F = 2.5$

$$\frac{\Delta L}{L} = 100 \mu\text{m}$$

$$= 100 \times 10^{-6}$$

$$\text{So, } \Delta R = G.F \times \frac{\Delta L}{L} \times R$$



$$\begin{aligned}
 &= 2.5 \times 100 \times 10^{-6} \times 600 \\
 &= 0.15 \\
 \text{So, } V_B &= \frac{V_s}{4} \times \frac{\Delta R}{R + 2\Delta R} \\
 &= \frac{4}{4} \times \frac{0.15}{600 + 2(0.15)} \\
 &= 2.4987 \times 10^{-4} \\
 &\approx 250 \mu V
 \end{aligned}$$

Variable inductance transducer

The variable inductance transducers work upon one of the following three principles :

- Change of self inductance
- Change of mutual inductance.
- production of eddy currents

Based on change of self inductance:

- The self inductance of a coil,

$$L = N^2 / R$$

Where N = number of turns
R = reluctance of the magnetic circuit
- The reluctance of the magnetic circuit $R = l / \mu A$

$$\therefore \text{Inductance } L = N^2 \mu (A/l)$$

$$= N^2 \mu G \text{ -----(1)}$$

where μ = effective permeability of the medium in and around the coil

$G = A / l$ = geometric form factor

A = area of cross-section of coil

l = length of the coil

- It is clear from equation (1) that the variation in inductance may be caused by :
- Change in number of turns, N
- Change in geometric configurations, G.
- Change in permeability, μ

Based on change of Mutual Inductance :

- An inductive transducer working on the principle of variation of mutual inductance uses multiple coils.

- The mutual inductance between the two coils is $M = K \sqrt{L_1 L_2}$
 L_1 and L_2 = self-inductance of two coils
K = coefficient of coupling
- Thus mutual inductance between the coils can be varied by variation of self-inductances or the coefficient of coupling
- The mutual inductance can be converted into a self-inductance by connecting the coils in series
- The self-inductance of such an arrangement varies between $L_1 + L_2 - 2M$ to $L_1 + L_2 + 2M$ with one of the coils being stationary while the other is movable.

Air cored coils:

They can be operated at a higher carrier frequency because of absence of eddy current losses in air cores

Iron cored coils:

Advantages:

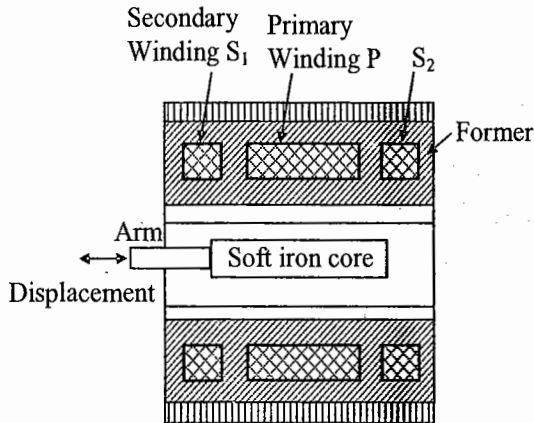
- Their size is much smaller than that air cored transducers on account of high permeability of iron cores.
- They are less likely to cause external magnetic fields and are less affected by stray magnetic fields

Based on production of Eddy currents:

- These inductive transducers work on the principle that if a conducting plate is placed near a coil carrying alternating current, eddy currents are produced in the conducting plate.
- The conducting plate acts as a short-circuited secondary winding of a transformer.
- The eddy currents flowing in the plate produce a magnetic field of their own which acts against the magnetic field production by the coil.
- This results in reduction of flux and thus the inductance of the coil is reduced.
- The nearer is the plate to the coil, the higher are the eddy currents and thus higher is the reduction in the inductance of the coil.

Linear Variable Differential Transformer (LVDT)

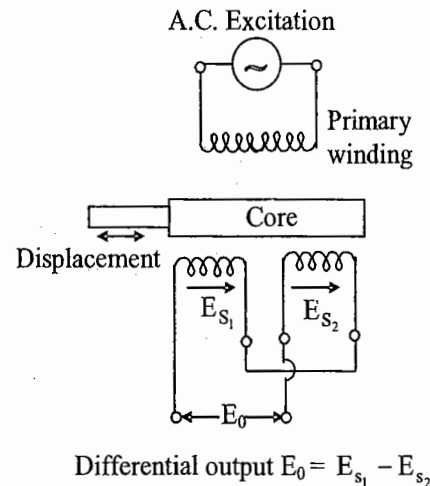
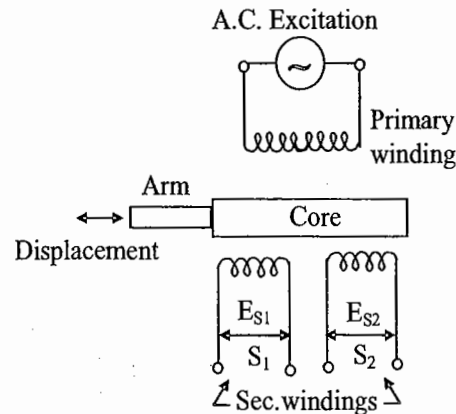
- The most widely used inductive transducer to translate the linear motion into electrical signals is the linear variable differential transformer (LVDT).
- The basic construction of LVDT is shown in figure



- The transformer consists of a single primary winding p and two secondary windings S_1 and S_2 wound on a cylindrical former.
- The secondary windings have equal number of turns and are placed identically on either side of the primary winding.
- The primary winding is connected to an alternating current source.
- A movable soft iron core is placed inside the former.
- The displacement to be measured is applied to the arm attached to the soft iron core.
- Since the primary winding is excited by an alternating current source, it produces an alternating magnetic field which in turn induces alternating current voltages in the two secondary windings.
- The output voltage of secondary S_1 is E_{S1} and that of secondary S_2 is E_{S2} .
- In order to convert the outputs from S_1 and S_2 into a single voltage signal, the two secondaries S_1 and S_2 are connected in series opposition.

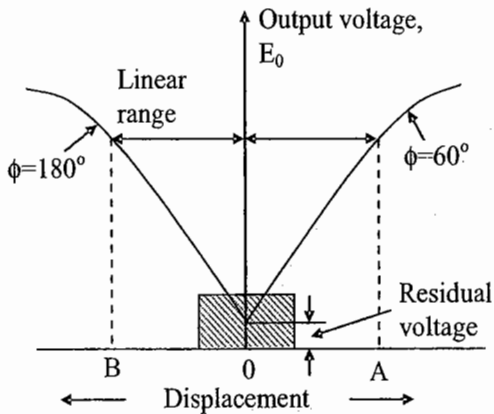
Thus the output of the transducer is the difference of the two voltages.

Differential output voltage, $E_o = E_{S1} - E_{S2}$



- When the core is at NULL position, the flux linking with both the secondary windings is equal and hence equal emfs are induced in them.
- Thus at NULL position, $E_{S1} = E_{S2}$ and hence the output voltage $E_o = 0$.
- If the core is moved to the left of the NULL position, more flux links with winding S_1 and less with winding S_2 .
- The output voltage E_{S1} of the secondary winding S_1 is more than E_{S2} , the output voltage of secondary winding S_2 .
- The magnitude of output voltage $E_o = E_{S1} - E_{S2}$ and is in phase with the primary voltage.

- If the core is moved to the right of the null position, the flux linking with winding S_2 becomes larger than that with winding S_1 .
- Thus $E_o = E_{S2} - E_{S1}$ which is 180° out of phase with primary voltage.
- The following figure shows the variations of output voltage against displacement for various positions of the core.



Advantages:

- High range
- Friction and Electrical Isolation
- Immunity from external effects
- High input and high sensitivity
- Ruggedness
- Low hysteresis
- Low power consumption

Disadvantages:

- Relatively large displacements are required for appreciable differential output.
- Sensitive to stray magnetic fields
- Temperature affects the performance of transducer.
- Dynamic response is limited mechanically by the mass of the core and electrically by the frequency response of the applied voltage.

Example 3:

The output voltage of a LVDT is 1.5V at maximum displacement. At a load of $0.5 \text{ M}\Omega$, the

deviation from linearity is maximum and it is $\pm 0.003 \text{ V}$ from a straight line through origin. the linearity at the given load is _____.

Ans: $\pm 0.2\%$

Sol: We know for LVDT

The percentage linearity

$$= \frac{\pm 0.003}{1.5} \times 100$$

$$= \pm 0.2\%$$

Example 4:

The output of the LVDT with input mechanical motion of 10 Hz and excitation frequency 400 Hz will contain frequencies.

- 10 Hz and 400 Hz
- 400 Hz only
- 10 Hz only
- 390 Hz and 410 Hz

Ans(d)

Sol: For LVDT

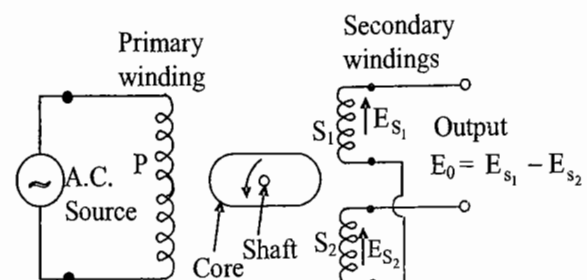
$f_{ex} \pm f_x$ is the modulated wave frequency
so modulated wave contains frequency

$$400 \text{ Hz} \pm 10 \text{ Hz}$$

$$= 390 \text{ Hz to } 410 \text{ Hz}$$

Rotary Variable Differential Transformer (RVDT)

- A variation of linear variable differential transformer (LVDT) which can be used to sense angular displacement is Rotary Variable Differential Transformer (RVDT).
- The circuit of RVDT is shown in the following figure.



- It is similar to LVDT except that its core is cam shaped and may be rotated between the windings by means of a shaft.
- The operation is similar to LVDT.
- Clockwise rotation produces an increasing voltage of a secondary winding of one phase while counter-clockwise rotation produces an increasing voltage of opposite phase.
- Hence, the amount of angular displacement and its direction may be ascertained from the magnitude and phase of the output voltage of the transducer.

Synchros

A synchro is an electromagnetic transducer which is commonly used to convert the angular position of the shaft into an electric signal.

- There are two types of synchro systems.
 - (i) Control or error detecting type
 - (ii) Torque transmission type

Resolvers

- Resolvers are used for conversion of angular position of a shaft into Cartesian coordinates.
- The output of the transducer is in the form of two signals, one proportional to the sine of the angle and the other proportional to cosine of the angle.
- Resolvers are classified into two groups
 - Computing resolvers
 - Synchro resolvers
- Computing resolvers are used for generating sine, cosine and tangent functions as well as for solving geometric relationships.
- Synchro resolvers are used for data transmission.

Applications:

- Vector resolution
- Vector composition
- Vector angle and component resolution
- Pulse amplitude control and pulse resolution

- Phase shifting

Capacitive transducer

The principle of operation of capacitive transducers is based upon the familiar equation for capacitance of a parallel plate capacitor.

$$\text{Capacitance, } C = \epsilon A/d$$

$$= \epsilon_r \epsilon_0 A/d$$

Where A = overlapping area of plates

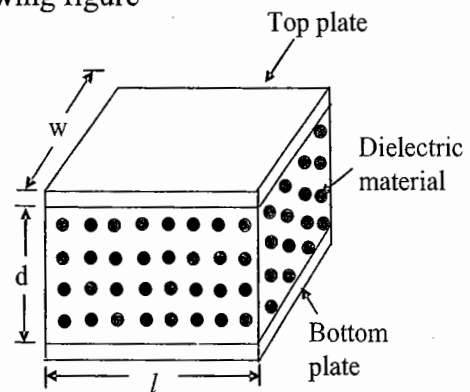
D = distance between two plates

$\epsilon = \epsilon_r \epsilon_0$ = permittivity of medium

ϵ_r = relative permittivity

ϵ_0 = permittivity of free space

A parallel plate capacitor is shown in the following figure



The capacitive transducer works on the principle of change of capacitance which may be caused by:

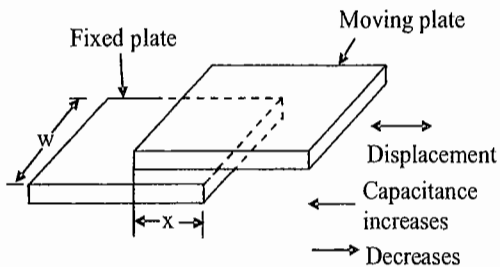
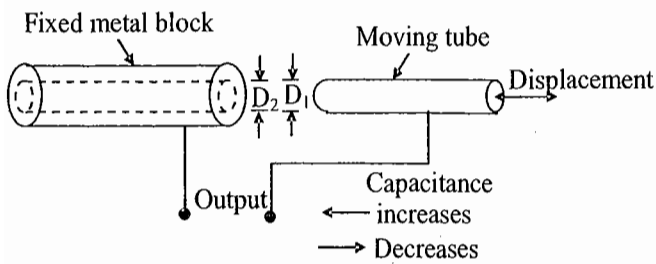
- Change in overlapping area A.
- Change in distance d between the plates.
- Change in dielectric constant

Change in Area of plates :

The equation for capacitance,

$$C = \epsilon A/d = \epsilon_r \epsilon_0 A/d$$

- The capacitance is directly proportional to the area A of the plates
- Thus the capacitance changes linearly with change in area of the plates.
- The elementary diagram of two types of capacitive transducers are shown in the following figures.



- For a parallel plate capacitor, the capacitance is

$$C = \epsilon A / d = \epsilon x w / d$$

Where x = length of overlapping part of plates

w = width of overlapping part of plates

$$\text{Sensitivity } S = \frac{\partial C}{\partial x} = \epsilon \frac{w}{d} \quad \text{F/m}$$

- This is suitable for measurement of linear displacements ranging from 1 mm to 10 mm
- Accuracy is as high as 0.005 %
- For a cylindrical capacitor, the capacitance is:

$$C = \frac{2\pi\epsilon x}{\log_e(D_2/D_1)} \quad \text{F}$$

Where x = length of overlapping part of cylinders

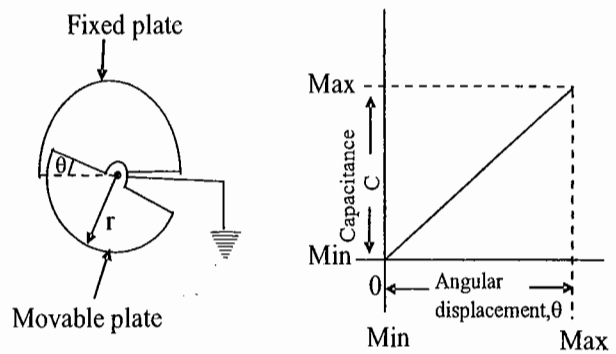
D_2 = inner diameter of outer cylindrical electrode

D_1 = outer diameter of inner cylindrical electrode

- Sensitivity $S = \frac{\partial C}{\partial x} = \frac{2\pi\epsilon}{\log_e(D_2/D_1)} \quad \text{F/m}$

Measurement of angular displacement:

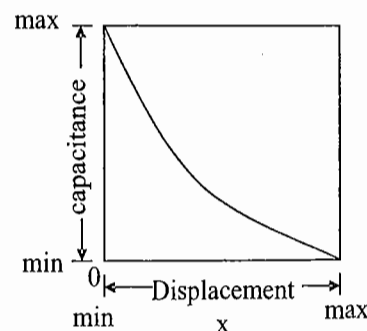
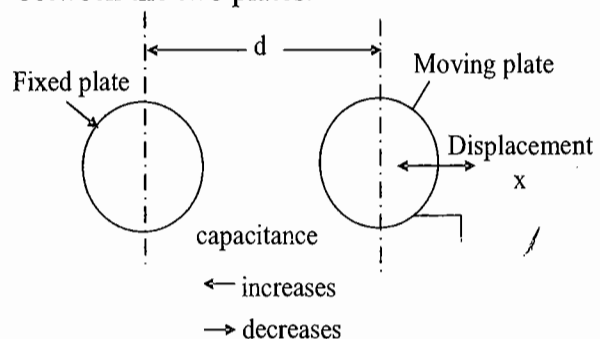
The following figure shows a two plate capacitor used for measurement of angular displacement.



- One plate is fixed and other is movable.
- The angular displacement to be measured is applied to the movable plate which changes the effective area between the plates and thus changes the capacitance.
- Maximum value of capacitance
Capacitance at an angle θ is
Where θ = angular displacement in radian.
- Sensitivity $S = \frac{\partial C}{\partial \theta} = \frac{\epsilon r^2}{2d}$
- The variation of capacitance with angular displacement is linear.

Change in distance between the plates:

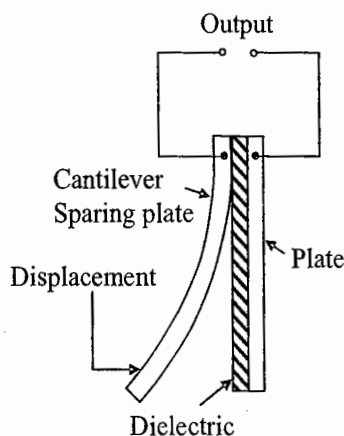
- The following figure shows the basic form of a capacitive transducer utilizing the effect of change of capacitance with change in distance between the two plates.



- One is fixed plate and the displacement to be measured is applied to the other plate which is movable.
- The capacitance C , varies inversely as the distance d between the plates, the response of the transducer is non-linear.

Capacitive transducer using cantilever spring plate:

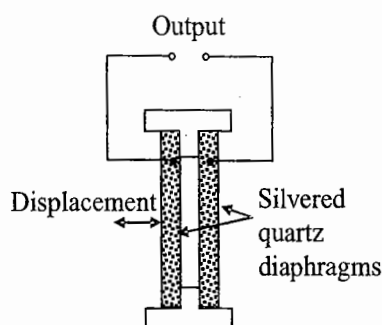
The following figure shows a capacitive transducer using cantilever spring plate



- The displacement when applied to the cantilever type spring plate moves it towards the second plate decreasing the distance.
- This increases the capacitance of the capacitor.

Capacitive transducer using Quartz diaphragms:

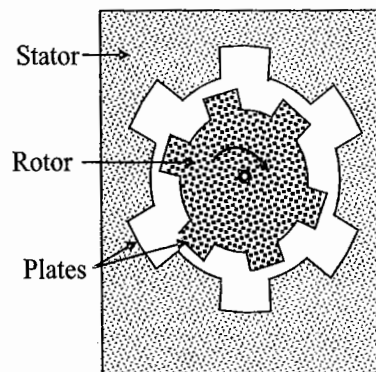
- The following figure shows a capacitive transducer using Quartz diaphragms.



- Two silvered quartz diaphragms form a capacitor
- The displacement of these diaphragms varies the capacitance of this capacitive transducer.

Rotational displacement measurement :

- The following figure shows an arrangement for rotational displacement measurement.



- As the rotor plates of the capacitor are displaced in the counter clockwise direction the capacitance increase
- The change in capacitance is a measure of the angular displacement.
- Effectively used for the measurement of torque.

Advantages:

- They require extremely small forces to operate them and hence are very useful for use in small systems.
- They are extremely sensitive
- They have a good frequency response. This response is as high as 50 kHz and hence they are very useful for dynamic studies.
- They have a high input impedance and therefore the loading effects are minimum.
- A resolution of the order of 2.5×10^{-3} mm can be obtained with these transducers.
- The capacitive transducers can be used for applications where stray magnetic fields render the inductive transducers useless.



- The force requirements of capacitive transducers is very small and therefore they require small power to operate them.

Disadvantages:

- The metallic parts of the capacitive transducers must be insulated from each other. In order to reduce the effects of stray capacitances, the frames must be earthed.
- The capacitive transducers show non-linear behaviour many a times on account of edge effects. Therefore guard rings must be used to eliminate this effect. Guard rings are also a must in order to eliminate the effect of stray electric fields, especially when the transducers have a low value of capacitance of the order of pF.
- The cable connecting the transducer to the measuring point is also a source of error. The cable may be source of loading resulting loss of sensitivity.

Applications:

- Used for the measurement of both linear and angular displacements.
- Used for the measurement of force and pressure. The force and pressure to be measured are first converted to displacement which causes a change of capacitance.
- Used directly as pressure transducers in all those cases where the dielectric constant of a medium changes with pressure.
- Used for measurement of humidity in gases since the dielectric constant of gases changes with change in humidity thereby producing a change in capacitance.
- Commonly used in conjunction with mechanical modifiers for measurement of volume, density, liquid, level, weight etc.